

TrES-3b

R-2811

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140721 · created 2026-09-05T01:24:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456859.6729 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.033 +/- 0.056
Transit depth (Rp/Rs)^2	0.11 +/- 0.37 [%]
Orbital Inclination (inc)	79.07 +/- 2.87
Airmass coefficient 1 (a1)	0.044 +/- 0.023
Airmass coefficient 2 (a2)	3.0 +/- 1.5
Scatter in the residuals of the lightcurve fit is	48.67 %
Best Comparison Star	#2 - [65, 32]
Optimal Aperture	2.4
Optimal Annulus	12.52
Transit Duration (day)	0.011 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.033 $\pm$ 0.056 vs 0.16309 (deviation 2.06 σ)
Duration fit vs published	0.011 d vs 0.0567 d (deviation 2.39 σ)
Mid-time O-C	-3.0 min
Depth SNR	0.5
Residual scatter	48.67 %

Light curve

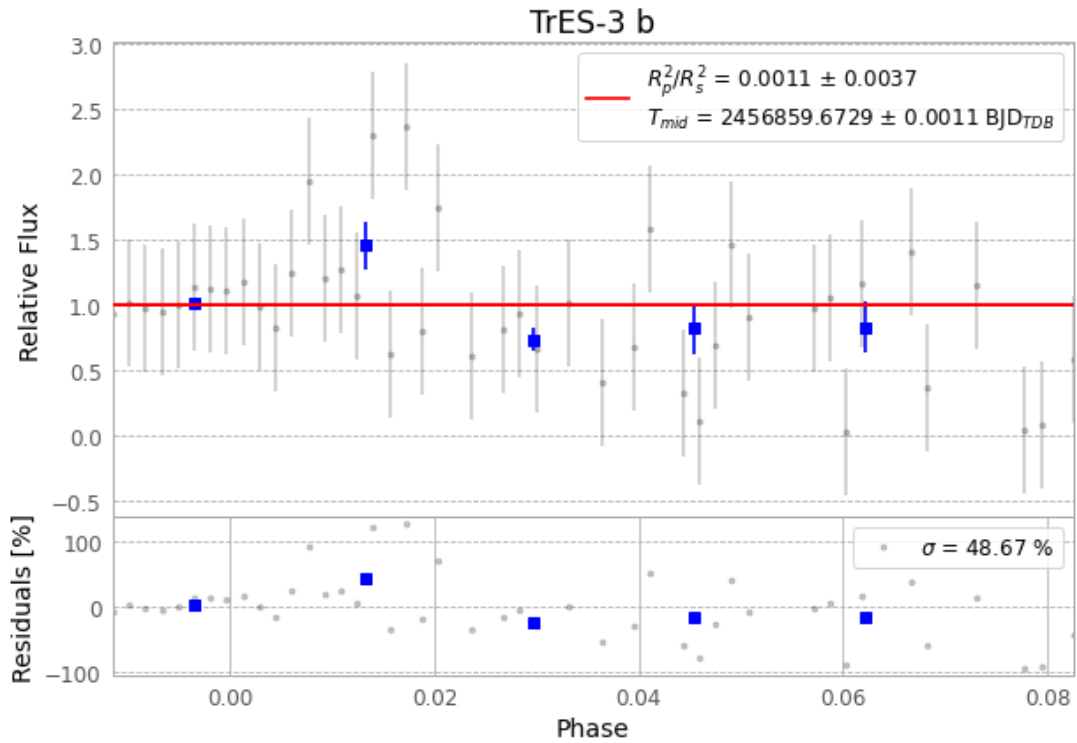


Figure 1 — FinalLightCurve_TrES-3 b_2014-07-20.png (SHA-256 acd1e6ceaaf92728...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 267cfc6d9b433e26...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), TRES-3140721034213.FITS → TRES-3140721064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140721.log (c79ee00ab6db...), FinalLightCurve_TrES-3 b_2014-07-20.png (acd1e6ceaaf9...), FinalLightCurve_TrES-3 b_2014-07-20.csv (173ad275e4cf...)

Compute resources

Wall time 447.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:24Z.